TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	78.0 ± 3.00
Correlation Stability	86.3 ± 3.20
Distribution Consistency	84.9 ± 1.81
Entropy Preservation	88.3 ± 1.19
Mean Z Score Drift	0.2 ± 0.04
Categorical Fidelity	83.5 ± 0.67
Privacy Risk	49.8 ± 1.54
Data Density	99.9 ± 0.13
Coverage Score	19.0 ± 5.64
Feature Accuracy	78.2 ± 1.89
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	42.140	-2.606	5.8%	OK
TB	3.299	1.900	-1.399	42.4%	Drift
DB	1.486	0.786	-0.700	47.1%	Drift
Alkphos	290.576	240.218	-50.358	17.3%	Warn
Sgpt	80.714	45.943	-34.771	43.1%	Drift
Sgot	109.911	46.596	-63.314	57.6%	Drift
TP	6.483	6.900	+0.417	6.4%	OK
ALB	3.142	3.496	+0.354	11.3%	Warn
A/G Ratio	0.947	1.009	+0.062	6.5%	OK
Selector	1.286	1.392	+0.106	8.2%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	78.0 ± 3.00
Correlation Stability (*2)	86.3 ± 3.20
Distribution Consistency (*3)	84.9 ± 1.81
Entropy Preservation (*4)	88.3 ± 1.19
Mean Z Score Drift (*5)	0.2 ± 0.04
Categorical Fidelity (*6)	83.5 ± 0.67
Privacy Risk (*7)	49.8 ± 1.54
Data Density (*8)	99.9 ± 0.13
Coverage Score (*9)	19.0 ± 5.64

Feature Accuracy (*10)	78.2 ± 1.89
Novelty Score (*11)	0.4 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.6059829059829059	0.7068376068376068	0.00341880341880341	
F1	0.8108734553482932	0.6934398299316046	0.8039156496383916	-0.00695780570990162	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7119658119658119	0.594017094017094	0.6871794871794872	-0.0247863247863248	
F1	0.8196135034430998	0.6803501856962714	0.786712691733162	-0.032900811709937736	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6478632478632479	0.7051282051282052	0.03076923076923077	*
F1	0.7821343746683586	0.7307522859080041	0.8039948713265893	0.021860496658230588	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6786324786324787	0.6820512820512821	-0.00427350427350425 8	
F1	0.7971299319768493	0.7788830942874784	0.7903182388309042	-0.006811693145945219	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7094017094017094	0.6290598290598289	0.7085470085470086	-0.00085470085470086 27	
F1	0.8116674386592997	0.7125418367306569	0.8080729872203198	-0.003594451438979751 2	

Model: NB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.6230769230769231	0.5743589743589743	0.015384615384615396	*
F1	0.5587432119380917	0.6850389424158789	0.5857327842664896	0.026989572328397948	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6478632478632478	0.535042735042735	0.6641025641025642	0.016239316239316248	
F1	0.7534952885557236	0.6228991986067682	0.7583646298941948	0.004869341338471345	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6863247863247863	0.7068376068376068	0	
F1	0.8279091624529537	0.7869849882605229	0.8279091624529537	0	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5991452991452991	0.7102564102564102	0.007692307692307676	
F1	0.7997903571549948	0.6878282005706811	0.8014918107851701	0.0017014536301753203	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6803418803418804	0.6025641025641025	0.6957264957264958	0.015384615384615408	
F1	0.7870498263126752	0.6847270326435153	0.794434036790191	0.007384210477515851	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4
2	27	Male	1.2	0.4	179	63	39	6.1	3.3	1.1
3	29	Male	1.2	0.4	160	20	22	6.2	3.0	0.9
4	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
5	55	Male	0.9	0.2	190	25	28	5.9	2.7	0.8
6	35	Male	1.8	0.6	275	48	178	6.5	3.2	0.9
7	30	Female	1.0	0.4	200	50	30	6.5	3.2	1.0
8	45	Male	0.8	0.3	150	40	20	7.0	3.5	1.2
9	52	Female	1.5	0.7	250	60	35	6.0	3.0	0.9
10	29	Male	0.9	0.2	180	30	25	5.8	2.8	1.1
11	38	Female	2.1	1.0	300	70	40	7.2	3.6	1.5
12	60	Male	0.7	0.1	220	20	15	6.3	2.9	0.8
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	46	Female	1.4	0.4	298	509	623	3.6	1.0	0.3
2	55	Male	18.4	8.8	206	64	178	6.2	1.8	0.4
3	47	Female	3.0	1.5	292	64	67	5.6	1.8	0.47
4	21	Male	1.0	0.3	142	27	21	6.4	3.5	1.2
5	52	Male	0.6	0.1	171	22	16	6.6	3.6	1.2
6	46	Male	20.0	10.0	254	140	540	5.4	3.0	1.2
7	34	Female	0.7	0.2	210	30	25	6.5	3.0	0.9
8	29	Male	1.2	0.4	180	40	30	7.0	4.0	1.1
9	50	Female	2.5	1.0	300	50	60	6.8	3.5	0.7
10	45	Male	1.8	0.5	250	20	45	7.2	3.2	1.0
11	60	Female	3.0	1.5	400	70	80	7.5	4.5	0.8
12	38	Male	0.9	0.3	220	25	35	6.2	2.8	1.2
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

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B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	1.0	0.3	154	38	21	6.8	3.9	1.3
2	41	Male	1.2	0.5	246	34	42	6.9	3.4	0.97
3	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
4	60	Male	5.8	2.7	599	43	66	5.4	1.8	0.5
5	22	Female	1.1	0.3	138	14	21	7.0	3.8	1.1
6	42	Female	0.8	0.2	195	18	15	6.7	3.0	0.8
7	34	Female	0.9	0.4	150	30	25	6.5	3.2	1.0
8	45	Male	1.5	0.6	200	40	30	7.0	4.0	1.2
9	29	Female	0.8	0.3	180	20	22	6.8	3.5	1.1
10	52	Male	2.1	0.9	250	50	40	7.5	4.2	1.3
11	39	Female	1.0	0.4	160	25	27	6.9	3.8	1.2
12	41	Male	3.0	1.2	300	60	50	8.0	4.5	1.5
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	0.8	0.2	127	29	30	4.9	2.7	1.2
2	21	Male	1.0	0.3	142	27	21	6.4	3.5	1.2
3	22	Male	0.8	0.2	300	57	40	7.9	3.8	0.9
4	34	Female	0.6	0.1	161	15	19	6.6	3.4	1.0
5	60	Male	11.0	4.9	750	140	350	5.5	2.1	0.6
6	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
7	29	Female	0.9	0.3	150	22	30	6.5	3.2	1.1
8	38	Male	1.2	0.5	200	45	50	7.0	3.8	0.9
9	45	Female	0.7	0.2	180	30	40	6.0	3.0	1.2
10	52	Male	1.5	0.6	220	50	55	6.8	3.5	1.0
11	60	Female	2.0	1.0	250	60	70	7.5	4.0	1.3
12	34	Male	0.8	0.3	160	25	35	6.2	3.1	1.1
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	66	Male	0.6	0.2	100	17	148	5.0	3.3	1.9
2	75	Male	10.6	5.0	562	37	29	5.1	1.8	0.5
3	65	Male	0.7	0.2	265	30	28	5.2	1.8	0.52
4	32	Male	23.0	11.3	300	482	275	7.1	3.5	0.9
5	60	Male	8.6	4.0	298	412	850	7.4	3.0	0.6
6	70	Male	1.7	0.5	400	56	44	5.7	3.1	1.1
7	29	Female	1.2	0.4	180	22	25	6.5	3.1	1.1
8	54	Male	2.1	1.0	210	40	35	7.0	3.5	0.9
9	38	Female	0.9	0.3	150	30	28	6.2	3.0	1.3
10	45	Male	1.5	0.7	250	50	45	6.8	3.4	1.0
11	60	Female	0.8	0.2	190	25	30	7.1	3.2	1.2
12	33	Male	3.0	1.2	220	55	60	6.7	3.6	0.8
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	0.7	0.2	174	32	14	7.8	4.2	1.1
2	22	Male	2.7	1.0	160	82	127	5.5	3.1	1.2
3	73	Male	1.9	0.7	1750	102	141	5.5	2.0	0.5
4	65	Male	0.7	0.2	265	30	28	5.2	1.8	0.52
5	42	Male	11.1	6.1	214	60	186	6.9	2.8	2.8
6	7	Male	0.5	0.1	352	28	51	7.9	4.2	1.1
7	34	Female	1.2	0.5	210	45	50	6.5	3.2	1.0
8	29	Male	0.9	0.3	180	25	30	5.8	2.9	0.9
9	55	Female	2.5	1.2	300	60	70	7.0	3.5	1.1
10	47	Male	1.8	0.7	250	40	45	6.2	3.0	0.8
11	62	Female	3.1	1.0	400	80	90	7.5	3.8	1.2
12	38	Male	0.7	0.2	220	35	40	6.0	2.8	0.7
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
2	18	Male	0.8	0.2	282	72	140	5.5	2.5	0.8
3	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
4	50	Male	7.3	3.7	92	44	236	6.8	1.6	0.3
5	40	Male	0.7	0.2	176	28	43	5.3	2.4	0.8
6	10	Female	0.8	0.1	395	25	75	7.6	3.6	0.9
7	34	Female	1.2	0.4	210	30	45	6.0	3.0	1.0
8	45	Male	0.9	0.3	180	25	35	6.5	3.5	1.2
9	29	Female	0.8	0.2	150	20	30	5.8	2.8	0.9
10	52	Male	2.5	0.7	220	40	50	7.2	3.8	1.1
11	60	Male	3.0	1.0	250	50	60	6.9	4.0	1.3
12	38	Female	1.5	0.5	190	35	55	6.3	3.2	1.0
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	52	Male	1.8	0.8	97	85	78	6.4	2.7	0.7
2	66	Male	0.8	0.2	165	22	32	4.4	2.0	0.8
3	32	Female	0.7	0.1	240	12	15	7.0	3.0	0.7
4	70	Male	1.7	0.5	400	56	44	5.7	3.1	1.1
5	46	Female	0.8	0.2	182	20	40	6.0	2.9	0.9
6	50	Male	0.9	0.3	901	23	17	6.2	3.5	1.2
7	29	Female	0.6	0.2	150	30	25	6.5	3.2	0.8
8	54	Male	1.2	0.4	220	45	50	7.0	3.8	1.0
9	37	Female	0.9	0.3	180	35	40	6.8	3.6	0.9
10	63	Male	2.5	1.0	300	60	70	7.5	4.0	1.1
11	22	Female	0.8	0.2	140	20	30	6.2	3.1	0.7
12	48	Male	1.6	0.5	250	55	65	7.2	3.9	1.2
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
2	50	Male	1.1	0.3	175	20	19	7.1	4.5	1.7
3	42	Male	16.4	8.9	245	56	87	5.4	2.0	0.5
4	45	Male	3.2	1.4	512	50	58	6.0	2.7	0.8
5	66	Male	0.8	0.2	165	22	32	4.4	2.0	0.8
6	30	Female	0.7	0.2	63	31	27	5.8	3.4	1.4
7	29	Female	1.2	0.4	150	30	25	6.5	3.2	1.1
8	47	Male	3.5	1.5	220	45	60	7.0	4.0	1.3
9	36	Female	0.9	0.2	180	25	30	6.8	3.5	1.0
10	52	Male	4.2	2.1	300	50	70	7.5	4.2	1.2
11	41	Female	1.5	0.6	210	35	40	6.9	3.8	1.1
12	34	Male	2.0	0.8	160	28	33	6.7	3.4	1.0
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	27	Male	1.3	0.6	106	25	54	8.5	4.8	nan
2	55	Female	0.8	0.2	155	21	17	6.9	3.8	1.4
3	36	Male	5.3	2.3	145	32	92	5.1	2.6	1.0
4	26	Male	42.8	19.7	390	75	138	7.5	2.6	0.5
5	52	Male	0.6	0.1	171	22	16	6.6	3.6	1.2
6	33	Male	0.7	0.2	256	21	30	8.5	3.9	0.8
7	34	Female	1.2	0.4	150	30	40	7.5	3.8	1.2
8	29	Male	0.9	0.3	200	25	35	6.8	3.5	1.0
9	45	Female	2.5	1.0	180	22	28	7.2	3.6	1.1
10	52	Male	0.8	0.2	160	40	50	8.0	4.0	1.0
11	39	Female	3.1	1.5	190	35	45	7.0	3.9	1.3
12	27	Male	1.5	0.5	210	20	30	6.5	3.4	1.1
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3